

Users, Requirements, and Constraints

Software Engineering

Enough shared understanding to build something useful

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Enough understanding to begin

“The perfect is the enemy of the good.”

Agree on a useful next step.

Keep open questions visible.

Who depends on the result?

Person	What matters to them
Volunteer	Am I confirmed?
Organizer	Which shifts are uncovered?
Treasurer	What will this cost?
Next year's officers	Can we keep it running?

Watch the work

Approach

Ask about an actual event

Watch the work

Try a rough sketch

Starting point

“Tell me about the last cancellation.”

“Show me how you update the schedule.”

“Where do I approve them?”

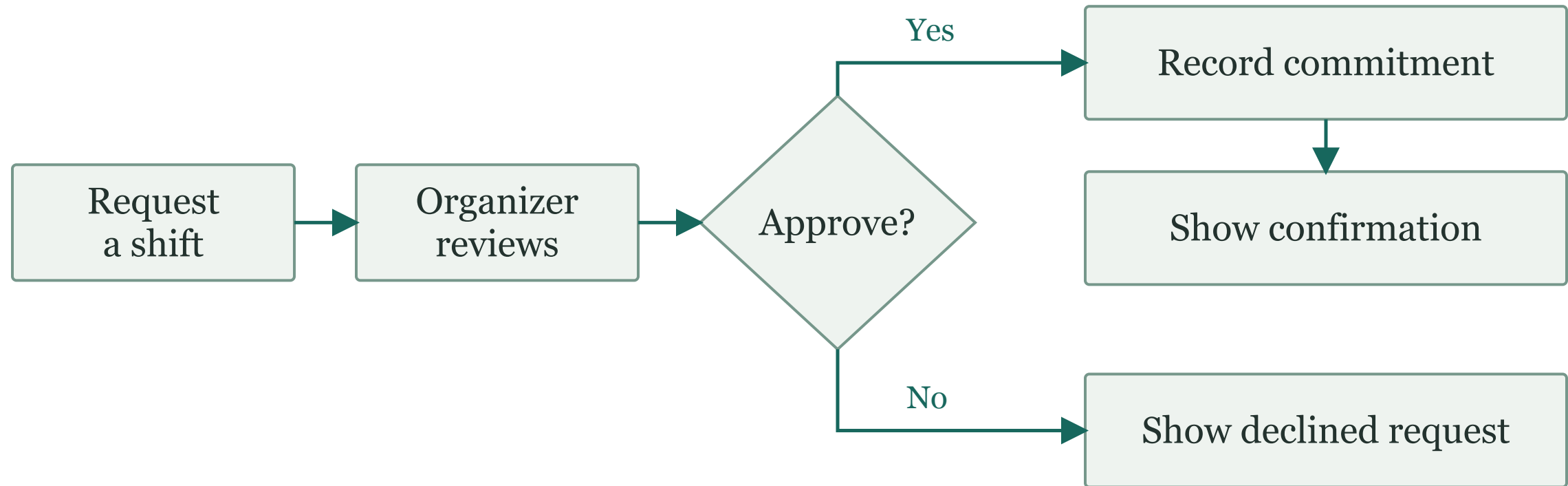
What did the first description leave out?

Need, requirement, design choice

Level	Example
Need	“I need to know I’m on the schedule.”
Requirement	View my confirmed event and shift time
Design choice	A green checkmark beside my name

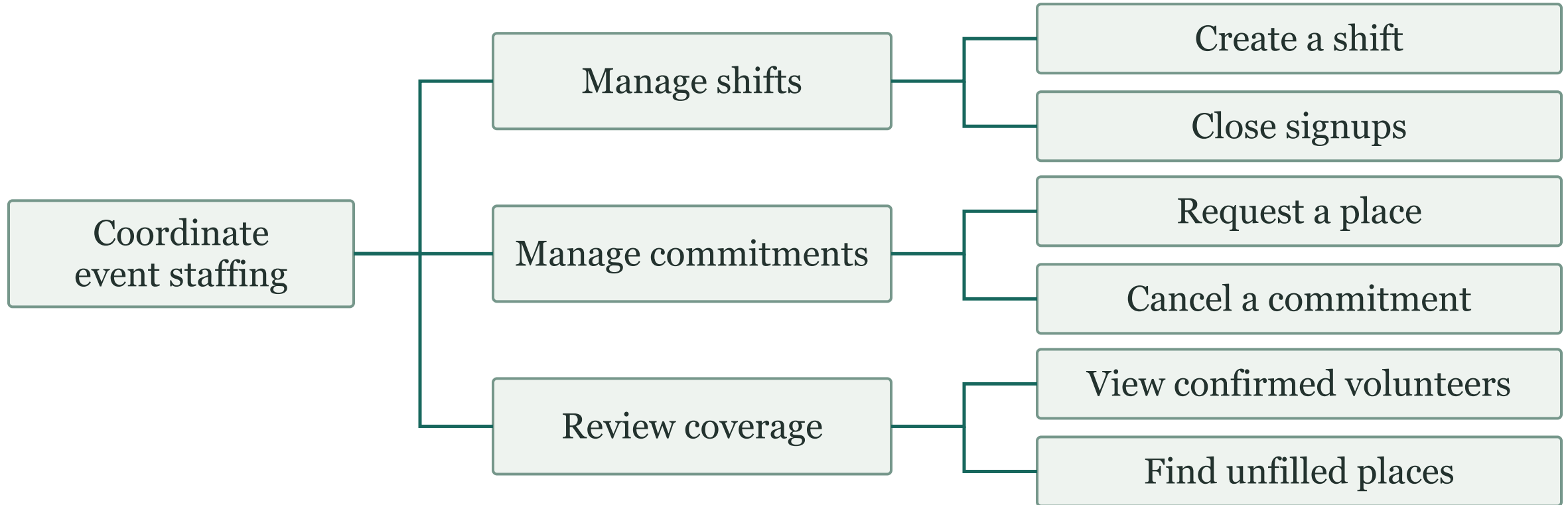
What does the checkmark promise?

A proposed signup workflow



Does a request reserve a place?

A function hierarchy



Lines mean “includes,” not “happens next.”

What makes it a car?

Basic function

Move people from place to place

Safety constraint

Must not catch fire in a collision at parking-lot speed

It must be a car, but it also must be an acceptable car.

Children's pajamas

Must not be flammable.

Functional or nonfunctional?

Important or not important?

Function and constraint

Product	Function	Constraint
Candy bar	Food to eat	No mold
Nerf gun	Fire foam darts	Below a specified velocity
Rocket	Carry a payload into flight	No toxic fuel dropped on the ground

Performing the function is only part of meeting the requirements.

Make quality concrete

Vague request

A useful question

Reliable

What survives a restart?

Private

Who may see contact details?

Easy to use

Can a new member sign up without help?

Find out what acceptable means here.

Say exactly what you mean

“Automatically notify everyone whenever anything changes.”

A proposal: when a confirmed volunteer cancels, email the event organizer immediately.

Who needs to know what, and when?

When needs conflict

Volunteer

“My plans can change.”

Organizer

“I need dependable coverage.”

What should happen to a late cancellation?

What can help before the event?

The event is in **ten days**.

The budget is **zero**.

Which useful part can be ready?

Checking the result

Validation

Does this solve the actual problem?

Volunteers know whether they are confirmed

Verification

Does it behave as specified?

Cancellation follows the agreed rule

A correctly built request form can still leave everyone uncertain.

Try the cancellation question

“People should be able to cancel, but I can’t have everyone disappearing just before the event.”

Propose a rule. Mark it as a proposal.

What example would reveal a disagreement?

Make the next commitment clear

Agree on what the next useful piece should do.

Record the questions that remain open.

Leave people a fair chance to finish work they can be proud of.